

ARJUN SUBHASH

arjunsbhash33@gmail.com | LinkedIn | GitHub | +91 7593965399

EDUCATION

- **Central University of Tamil Nadu, Thiruvavur, India** *Nov 2021 – Jun 2026*
Integrated MSc. Chemistry
CGPA: 8.5/10
- **Aravinda Vidyamandiram, Kottayam, India** *May 2017 – Apr 2019*
Senior Secondary Education
CBSE (84%)
- **Trinity Lyceum, Kollam, India** *Jun 2005 – Apr 2017*
Matriculation
ICSE (94%)

RESEARCH EXPERIENCE

- **Computer aided solvent design for enhanced guanidine crystallization in post combustion CO₂ capture** *Sep 2026 – Present*
Project Associate-I, CSIR-IICT
PI: **Dr. Mohammad Shahab**
 - Investigation of solvent-dependent CO₂ capture and guanidine-based crystallization using DFT and machine-learning-assisted molecular design.
- **Machine Learning Guided Computational Investigation of Cu^(I)-NHC Complexes for CO₂ Insertion** (Project Report) *Jan 2026 – Jun 2026*
M.Sc. Project, Central University of Tamil Nadu
PI: **Dr. Prabha Vadivelu**
 - Investigated CO₂ insertion into a library of 36 Cu^(I)-NHC complexes using DFT to establish structure–activity relationships governing catalytic reactivity.
 - Developed descriptor-based machine learning models to predict CO₂ insertion barriers and enable computational catalyst screening.
- **A Theoretical TD-DFT Investigation of Thermally Activated Delayed Fluorescence (TADF) Compounds Selected using Data-Driven Models** (Certificate) *Jun 2025 – Aug 2025*
Summer Intern, Digital University of Kerala
PI: **Prof. Dr. Anoop Ayyappan**
 - Performed TD-DFT calculations and electronic structure analyses of donor–acceptor TADF systems using frontier molecular orbitals, singlet–triplet energy gaps, oscillator strengths, and quantitative hole–electron descriptors to investigate excited-state properties and charge-transfer behaviour.
- **Tuning the Reactivity of N-Heterocycles and N-Heterocyclic Carbenes via SN₂ Mechanism: A Computational Study** (Certificate) *May 2024 – Jul 2024*
Summer Intern, SRIBS
PI: **Prof. Dr. Suresh C H**
 - Performed DFT studies on SN₂ reaction pathways of N-heterocyclic systems to understand reaction mechanisms, activation barriers, and substituent effects.
- **Comprehensive Study of Mineral Extraction and Processing of Titanium Dioxide** (Certificate) *Jun 2023 – Jul 2023*
Summer Intern, The Kerala Minerals & Metals Ltd.
PI: **Dr. Gireesh V S**
 - Gained industrial exposure to chloride-route TiO₂ production and quality control practices in a large-scale chemical manufacturing environment.

PUBLICATIONS

- Giri, S., Banerjee, M., **Subhash, A.**, Rajesh, S., Padinjarethil, S. E., & Anoop, A. (2026). Chemical Space Exploration and Molecular Design of Organic Donor–Acceptor TADF Emitters. *Journal of Materials Chemistry C*. <https://doi.org/10.1039/D6TC00707D>
- Vijayan, A., **Subhash, A.**, Dash, B. R., Mishra, A. K. & Prakash, J. Direct Quantification of L-Kynurenine in Urine by Angular Sweep Total Fluorescence Spectroscopy Coupled with MCR-ALS. (Manuscript under Review)
- Subhash, A.**, Sivaramakrishnan, M. P., & Vadivelu, P. A Theoretical Investigation of Cu^(I)–NHC Complexes for CO₂ Insertion. (Manuscript under Preparation)

CONFERENCES

- Arjun Subhash**; Prabha Vadivelu. "Machine Learning Guided Computational Investigation of Cu^(I)-NHC Complexes for Efficient CO₂ Insertion", New Trends in Applied Chemistry (NTAC 2026), Sacred Heart College (Autonomous), Thevara, 20th & 21st February, 2026. (Certificate)
- Arjun Subhash**; Anoop Ayyappan. "A Theoretical TD-DFT Investigation of Thermally Activated Delayed Fluorescence (TADF) Compounds Selected using Data Driven Models", Two Days National Level Conference on Advances in Computational Techniques for Physics, Chemistry & Mathematics (ACTPCM 2025), Sri Akilandeswari Women's College, Wandiwash, 8th & 9th October. ISBN: 978-81-993168-5-0 (Certificate)

SKILLS & TECHNIQUES

- Computational Chemistry:** DFT, TD-DFT, molecular docking and basic machine learning.
Software: Gaussian, ORCA, xTB, CREST, Multiwfn, AIMAll, AutoDock Vina, PyMOL, Chemcraft, ChemOffice, Open Babel.
- Experimental Techniques:** UV–Visible spectroscopy, IR spectroscopy, MP-AES Spectrometer, XRD, XRF, Particle size analysis and Pigment quality assessment.

TEACHING EXPERIENCE

- Chemistry Faculty, Epsilon - The Learning Hub** *Aug 2026 – Present*
Delivering foundational and higher secondary-level chemistry lectures aligned with CBSE curriculum.
- Chemistry Faculty, E-Care Classes** *Oct 2024 – Present*
Delivered over 300 hours of higher secondary-level chemistry lectures aligned with CBSE curriculum and competitive exam requirements.
- Chemistry Faculty, Bytesor** *Feb 2022 – Jan 2023*
Conducted over 50 hours of interactive online Chemistry sessions for High School students by simplifying complex concepts, guiding problem-solving and enhancing exam readiness.

ACHIEVEMENTS

- Qualified **GATE-2025** ; AIR-2619 (Scorecard)
- Qualified JEE(Main), NEET, ICAR & CUCET

EXTRA-CURRICULAR

- Three-time recipient of Form-3 certificate in Badminton (Certificate)
- Medalist at District Level Tamil Nadu Chief Minister's Trophy:
First Runner Up – Badminton Singles – 2023
Second Runner Up – Badminton Doubles – 2024
- Organizer and Technical Assistant at Spargere - 2025 (Department Science Day Celebration), Elixir - 2025 (Department Cultural) and numerous other university level programs.

REFERENCES

- **Dr. Prabha Vadivelu**
Assistant Professor
Central University of Tamil Nadu, India.
Email: prabhav@cutn.ac.in
 - **Dr. Anoop Ayyappan**
Professor
Digital University of Kerala, India.
Email: anoop.a@duk.ac.in
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